The Synchronous Field Theory and the Longitudinal Gamma-Ray Field Generalized Lorentz–Galilei Kinematics, the Variable-Coefficient Spherical Wave Equation, and the Casimir Boundary-Value Problem at V = 10⁹c; 28 Sep. 2026 Viorel Popescu

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The Synchronous Field Theory and the Longitudinal Gamma-Ray Field

Framework: Onduscular Theory (Teoria Onduscular, TauTimp) — Synchronous Field Theory (SFT) **Sources:** SynchronFieldTh.docx (14 Jan. 2026), Gamma.txt, Gamma_Academic_EN.md (Parts III–V)

Abstract

This paper restates the Synchronous Field Theory (SFT) in academic form. It replaces the provisional field equation of the original note with the variable-coefficient spherical wave equation for gamma radiation, $(R/r)\,\Delta H = V^{-2}\,\partial_t^2 H$. Gamma radiation is postulated to be a **longitudinal** wave propagating at the superluminal phase velocity $V = 10^9 c$ through an almost incompressible quantum space. Section 1 sets up the kinematics: a generalized Lorentz–Galilei transformation whose velocity parameter $\beta = V/c$ spans the Galilean ($\beta \ll 1$), Einsteinian ($\beta \to 1$) and synchronous ($\beta \gg 1$, imaginary Lorentz factor) regimes. The mechanical origin of the velocity is then discussed through the longitudinal sound-speed relation $V = \sqrt{\chi/\rho}$ for the compressibility of space, together with the primordial time quantum (PTQ) and the practical velocity limit. The core of the paper follows Parts III–V of the gamma-ray treatise. The field equation is separated, reduced exactly to the Airy equation and quantized by $H(2R,\theta,t)=0$. A two-boundary Casimir problem on $[2R,\ 2R_s]$ is solved in closed form, and the resulting longitudinal flux ($\propto r^{-4}$) and Casimir pressure ($\propto \hbar V/L^2$) are derived. All numerical values were recomputed independently. Where the sources disagree, the discrepancy is reported explicitly (Section 7).

1. Kinematic framework: the generalized Lorentz–Galilei transformation 1.1 The unified transformation

Let two inertial frames move with relative velocity V along x, and let $\beta = V/c$. The generalized transformation, which contains both the Lorentz and the Galilei transformations, is

$$ct' = \gamma\,(ct - \beta x), \qquad x' = \gamma\,(x - \beta ct), \qquad y' = y, \qquad z' = z, \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}}.$$

Equivalently, $t' = \gamma t - \gamma V x/c^2$ and $x' = \gamma x - \gamma V t$.

1.2 The three kinematic regimes

Regime	$\beta = V/c$	γ	Physical carrier
Galilean (classical)	$\beta \ll 1$	$\gamma \simeq 1$	massive bodies
Einsteinian (relativistic)	$\beta \to 1$	$\gamma \to \infty$	photon, graviton, magnetron, ν_e, ν_μ, ν_τ — **transverse** waves
Synchronous (SFT)	$\beta \gg 1$	γ purely imaginary	X-rays, γ-rays — **longitudinal** waves

For the synchronous regime with $V = 10^9 c$ one obtains

$$\gamma = \frac{1}{\sqrt{1-10^{18}}} = \frac{-i}{\sqrt{10^{18}-1}} \simeq -i \cdot 10^{-9}.$$

So the Lorentz factor becomes imaginary and very small in modulus. In the original note this regime is described as "β complex and very large". The imaginary factor is read as a reversal of the time ordering for zero-rest-mass carriers, in the sense of the negative-time particles of quantum information theory (QIT). In the limit $|\beta| \to \infty$ the SFT asserts that the local universe behaves as if it had no spatial or temporal separation: an event becomes **synchronous**, i.e. present throughout the local universe.

2. Longitudinal propagation and the compressibility of quantum space

For a longitudinal (compressional) wave in a continuous medium, the phase velocity is

$$V = \sqrt{\frac{\chi}{\rho}},$$

where χ is the compressibility (bulk) modulus and ρ the density. The SFT hypothesis is that quantum space is **almost incompressible** and of vanishing density ($\rho \to 0$). Eq. (2.1) then admits a very large propagation speed, bounded above by $\sim 10^{19}$ m/s. The original note uses the vacuum density $\rho = 9.9 \times 10^{-30}$ g cm^{-3} $= 9.9 \times 10^{-27}$ kg m^{-3} and the space compressibility modulus $\chi = 0.09070366748$ Pa, and quotes the resulting velocity as $V \sim 3 \times 10^{17}$ m/s $\simeq 10^9 c$ (see Section 7, item 1, for a numerical check of this step).

Gamma radiation is thus modelled as a compression wave of space itself rather than as a transverse oscillation of the electromagnetic field. This is the physical content of the postulate **"gamma radiation is longitudinal and propagates at $V = 10^9 c$"**, which is used throughout the paper.

3. Space quantum, primordial time quantum and practical velocity 3.1 Primordial time quantum

Light is assumed to be the fastest signal that moves *from one space quantum (SQ) to the next*. A gamma wave, by contrast, displaces each quantum only by one SQ, while the remaining quanta shift collectively in one direction during one primordial time quantum:

$$\mathrm{PTQ} = \frac{\mathrm{SQ}}{c} = 1.417 \times 10^{-27}\ \mathrm{s} \ \Rightarrow\ \mathrm{SQ} = c \cdot \mathrm{PTQ} = 4.248 \times 10^{-19}\ \mathrm{m}.$$

3.2 Carrier velocity vs. field velocity

The SFT distinguishes two velocities:

- the **carrier** (the gamma boson) moves between adjacent space quanta at c;
- the **field** (the torsion field it generates) propagates at $V = A\,c$, with $A = 10^9$.

The analogy is electrical conduction. Electrons drift through a copper lattice at $< 10^{-5}$ m/s, yet the electric field in the wire is transmitted at nearly c. This is possible for gamma quanta because they are indistinguishable bosons.

3.3 Practical velocity

The theoretical limit is $V = Ac \simeq 3 \times 10^{17}$ m/s. A physical object, however, is confined to the sphere of radius $R = 3.9228 \times 10^{22}$ m and acts over the time quantum $T_q = 1.765 \times 10^{-19}$ s. The practical velocity is therefore

$$V_p \lesssim (3\text{–}4) \times 10^{14}\ \mathrm{m/s} \ \approx\ 10^6 c.$$

4. The field equation of gamma radiation 4.1 From the provisional equation to the gamma equation

The original note starts from the one-dimensional provisional form

$$\frac{R_s}{r}\frac{\partial^2 f}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f}{\partial t^2},$$

with Dirichlet data $g(2R_s) = 0$ and $g'(0) = R$. In the present paper, Eq. (4.1) is **replaced** by the full spherical gamma-ray equation of Gamma.txt:

$$\boxed{\frac{R}{r}\,\Delta H(r,\theta,t)=\frac{1}{V^2}\,\frac{\partial^2 H(r,\theta,t)}{\partial t^2}},\qquad V=10^9 c.$$

Eq. (4.2) differs from (4.1) in two ways:

1. It uses the complete spherical Laplacian, including the geometric term $2/r\,\partial_r$.
2. Its anchor radius is R, the local-universe radius, instead of R_s.

The outer radius R_s is not discarded. It reappears as the second boundary of the Casimir problem in Part IV.

The factor R/r makes the effective squared speed position dependent, $V_{\text{eff}}^2(r)=RV^2/r$: the medium is "stiffer" near the origin and "softer" far from it. This is exactly the property that turns plane waves into Airy modes.

4.2 Physical constants

$$R=3.9228\times10^{22}\text{ m},\qquad R_s=7.7665\times10^{25}\text{ m},\qquad V=10^9c=2.99792\times10^{17}\text{ m/s}.$$

Part III — The variable-coefficient spherical wave equation III.1 Radial reduction

In the monopole sector (H independent of θ,φ) the Laplacian is $\Delta f=f_{rr}+2/r\,f_r$, and Eq. (4.2) becomes

$$\frac{R}{r}\left(\frac{\partial^2 f}{\partial r^2}+\frac{2}{r}\frac{\partial f}{\partial r}\right)=\frac{1}{V^2}\frac{\partial^2 f}{\partial t^2}.$$

III.2 Separation of variables

Substitute $f(r,t)=g(r)\,u(t)$. Since one side depends only on r and the other only on t, both must equal a constant, written $-\lambda^2$:

$$\frac{R}{r}\,\frac{g''+\frac{2}{r}g'}{g}=\frac{1}{V^2}\frac{u''}{u}=-\lambda^2,$$
$$u''+\lambda^2V^2u=0,$$
$$g''+\frac{2}{r}g'+\frac{\lambda^2}{R}\,r\,g=0.$$

The temporal solution is harmonic: $u(t)=A\cos(\omega t)+B\sin(\omega t)$, with $\omega=\lambda V$.

III.3 Exact reduction to the Airy equation

Substitute $g=v/r$. Then $g''+2/r\,g'=v''/r$ exactly, and (R) becomes $v''+\frac{\lambda^2}{R}\,r\,v=0$.

Next set $\xi=-\kappa r$ with $\kappa=(\lambda^2/R)^{1/3}$. This gives the standard Airy equation $v_{\xi\xi}-\xi v=0$, whose general solution is

$$g(r)=\frac{1}{r}\left[C_1\,\mathrm{Ai}(-\kappa r)+C_2\,\mathrm{Bi}(-\kappa r)\right].$$

III.4 Quantization from $H(2R,\theta,t)=0$

For the dominant branch we take $C_2=0$. The boundary condition $g(2R)=0$ then requires $\kappa_n\cdot 2R=|a_n|$, where a_n is the n-th zero of Ai. Hence

$$\boxed{\lambda_n=\frac{|a_n|^{3/2}}{2^{3/2}R},\qquad \omega_n=\lambda_nV=\frac{|a_n|^{3/2}\,V}{2^{3/2}R}}.$$

n	$\lvert a_n\rvert$	λ_n [m^{-1}]	ω_n [rad/s]	$T_n=2\pi/\omega_n$ [s]	T_n [days]
1	2.338107	3.222223×10^{-23}	9.659980×10^{-6}	6.504346×10^{5}	7.528
2	4.087949	7.449328×10^{-23}	2.233252×10^{-5}	2.813468×10^{5}	3.256
3	5.520560	1.169051×10^{-22}	3.504727×10^{-5}	1.792774×10^{5}	2.075
4	6.786708	1.593484×10^{-22}	4.777144×10^{-5}	1.315260×10^{5}	1.522
5	7.944134	2.018035×10^{-22}	6.049916×10^{-5}	1.038557×10^{5}	1.202

For comparison, the single-boundary problem $H(R)=0$ (file GammaRay_spherical_BC_R_TauTimp_EN.md) gives $\omega_n=V|a_n|^{3/2}/R$. Its fundamental period is $T_1=2.662$ days, and the ratio is $\omega_1^{(R)}/\omega_1^{(2R)}=2^{3/2}$, as predicted by the scaling $\omega_1\propto R_{bc}^{-3/2}$.

III.5 Remark on regularity at the origin

With $C_2=0$ the radial factor behaves like $g\simeq C_1\mathrm{Ai}(0)/r$ as $r\to0$. This is an integrable singularity of point-source type, not a regular solution. A strictly regular solution requires $v(0)=0$, which gives

$$v(r)\propto \mathrm{Bi}(0)\,\mathrm{Ai}(-\kappa r)-\mathrm{Ai}(0)\,\mathrm{Bi}(-\kappa r).$$

The quantization condition then becomes $\kappa_n\cdot 2R=z_n$, where z_n are the zeros of (III.6). Their numerical values are $z_n=2.666353,\ 4.342478,\ 5.741029,\ 6.986142,\ 8.128779$. The corresponding fundamental frequency is $\omega_1=1.176402\times10^{-5}$ rad/s, i.e. $T_1=6.182$ days, and the difference from the $C_2=0$ branch shrinks as n grows. For the regular solution $g'(0)=0$ identically. The original condition $g'(0)=R$, which belongs to the provisional equation (4.1) without the $2/r$ term, is therefore replaced by a normalization of the amplitude $g(0)$.

III.6 Complete field

In the monopole sector, $f(r,t)=\sum_{n=1}^{\infty}\frac{1}{r}\,\mathrm{Ai}\big(-(\lambda_n^2/R)^{1/3}r\big)[A_n\cos(\omega_n t)+B_n\sin(\omega_n t)]$.

Restoring the polar dependence through Legendre polynomials $P_\ell(\cos\theta)$ gives the full field:

$$\boxed{H(r,\theta,t)=\sum_{\ell=0}^{\infty}\sum_{n=1}^{\infty}\frac{1}{r}\,\mathrm{Ai}\left(-(\lambda_{n\ell}^2/R)^{1/3}r\right)P_\ell(\cos\theta)\left[A_{n\ell}\cos(\omega_{n\ell}t)+B_{n\ell}\sin(\omega_{n\ell}t)\right]},$$

with $\omega_{n\ell}=\lambda_{n\ell}V$. The centrifugal term $-\ell(\ell+1)/r^2$ is neglected at cosmological scale.

Part IV — The Casimir boundary-value problem on cosmological scales

IV.1 Reduced equation and boundary conditions

The static analysis between two widely separated spherical shells uses the 1+1 flat-space proxy of Eq. (4.2):

$$\frac{1}{V^2}\frac{\partial^2 f}{\partial t^2}-\frac{\partial^2 f}{\partial r^2}=0,$$

with

$$f(2R,0)=0,\qquad f(2R_s,0)=0,\qquad 2R=7.8456\times10^{22}\ \text{m},\quad 2R_s=1.5533\times10^{26}\ \text{m}.$$

In SFT language, $2R$ is the domain covered by a gamma ray within one primordial time unit (PTU), and $2R_s$ is the outer onduscular boundary that appeared in the original Eq. (4.1).

IV.2 Degree-two static ansatz

$$f(r,t)=\frac{(-4R^2+2Rr)\,V^2+(-4R^2+r^2)\,c_2}{8\pi r^2V^2}.$$

The first condition, $f(2R,0)=0$, is satisfied identically for every c_2. The second, $f(2R_s,0)=0$, gives

$$4R(R_s-R)V^2+4(R_s-R)(R_s+R)\,c_2=0\ \Rightarrow\ \boxed{c_2=-\frac{R\,V^2}{R_s+R}}.$$

IV.3 Numerical value for $V=10^9c$

With $V^2=8.98755\times10^{34}$ m² s⁻² and $R_s+R=7.77042\times10^{25}$ m,

$$\boxed{c_2=-4.5373\times10^{31}\ \text{m}^2\,\text{s}^{-2}}.$$

This value corrects the -4.5164×10^{31} quoted in the source; see Section 7, item 2.

IV.4 Explicit static solution

$$\boxed{f(r)=\frac{1}{8\pi r^2}\left[-\frac{4R^2R_s}{R_s+R}+2R\,r-\frac{R}{R_s+R}\,r^2\right]},$$

with numerator coefficients $N(r)=\alpha_0+\alpha_1 r+\alpha_2 r^2$:

Coefficient	Expression	Value
α_0	$-4R^2R_s/(R_s+R)$	-6.1522×10^{45} m²
α_1	$2R$	7.8456×10^{22} m
α_2	$-R/(R_s+R)$	-5.0484×10^{-4}

Direct substitution confirms $f(2R)=f(2R_s)=0$ to machine precision.

Part V — Longitudinal flux and Casimir radiation pressure

V.1 Longitudinal gamma-ray flux density

$$\Phi(r)=\frac{f(r)}{4\pi r^2}=\frac{\alpha_0+\alpha_1 r+\alpha_2 r^2}{32\pi^2r^4}.$$

At large r the flux falls off faster than the r^{-2} law of a transverse photon field. This is consistent with the longitudinal character of the gamma field: the energy is carried by compression of space along $\boldsymbol{k}$, not by a transverse oscillation spread over the sphere $4\pi r^2$.

V.2 Eigenfrequencies

The eigenfrequencies carried over from Part III are $\omega_n=|a_n|^{3/2}V/(2^{3/2}R)$, with $\omega_1=9.660\times10^{-6}$ rad/s.

V.3 Casimir radiation pressure

The zero-point stress of the longitudinal modes generates a Casimir-type pressure across the gap $L=2(R_s-R)=1.5525\times10^{26}$ m:

$$\mathcal{P}_{\text{Casimir}}=\frac{\hbar V}{2L}\sum_n\omega_n\,,\qquad F_{\text{Casimir}}=-\frac{\partial\mathcal{P}_{\text{Casimir}}}{\partial L}\propto\frac{\hbar V}{L^2}.$$

The effect keeps the $1/L^2$ form of the electromagnetic Casimir effect but is enhanced by the factor $V/c=10^9$. In the SFT this is the mechanical signature of the almost incompressible quantum space: the stiffer the medium, the denser the longitudinal mode spectrum, and the larger the vacuum pressure.

6. Synthesis within the Synchronous Field Theory

6.1 Characteristic time scales

Quantity	Definition	Value
Primordial time quantum	$\text{PTQ}=\text{SQ}/c$	1.417×10^{-27} s
Time quantum	T_q	1.765×10^{-19} s
Transit R at speed V	R/V	1.3085×10^{5} s (1.51 days)
Transit $2R$ at speed V (one PTU)*	$2R/V$	2.6170×10^{5} s (3.03 days)
Fundamental mode, BC $=R$	$2\pi R/(V\lvert a_1\rvert^{3/2})$	2.2996×10^{5} s (2.66 days)
Fundamental mode, BC $=2R$	$2\pi/\omega_1$	6.5043×10^{5} s (7.53 days)

*The identification of one PTU with the transit time $2R/V$ is our interpretation of the statement "the γ ray spreads within a 2R radius in one PTU".

The field thus crosses the local universe in a few days. This is the quantitative meaning of "synchronous": for any process slower than a few days, a gamma-field event is effectively present everywhere within radius $2R$ at once.

6.2 Dimensional classification of the synchronous particles

The original note classifies the particles of the SFT by their exponents with respect to mass (M), magnetic mass (M_g), energy (E), a gravitational-type dimension (G), length (L) and time (T):

Particle	M	M_g	E	G	L	T
Particle_{Bx}	0	1	0	1	0	-1
Particle_{E} (spherical)	0	0	1	1	0	-1
Particle γ (γ-ray, spherical)	0	0	0	1	1	-1
$\text{Particle}_{\text{Activate}}$ (entanglement)	0	0	0	-1	1	-1
$\text{Particle}_{\text{Synchronous}}$ (spherical)	0	0	0	1	-1	-1

Of these five particles, only the γ-ray is accessible to routine laboratory study. It is therefore the natural experimental probe of the SFT.

6.3 A proposed observational test

The original note proposes directing a γ-ray beam toward the CMB cold spot and superposing it on a visible-light beam. It predicts an interference pattern whose "delay between the two universes" exceeds $32\,1/2$ years, and whose appearance changes with the pointing direction. The present paper does not derive this figure. It is recorded as the author's prediction and remains to be linked quantitatively to the time scales of §6.1.

7. Consistency notes

1. **Compressibility velocity.** With $\rho = 9.9 \times 10^{-27}$ kg m⁻³, Eq. (2.1) gives $V = 1.005 \times 10^{13}\sqrt{\chi}$ m/s, and for $\chi = 0.0907$ Pa this is $V = 3.03 \times 10^{12}$ m/s. The source instead quotes the prefactor 1.054×10^{17} and the result 3×10^{17} m/s. Obtaining $V = 3 \times 10^{17}$ m/s from $\chi = 0.0907$ Pa requires an effective density $\rho \simeq 1.0 \times 10^{-36}$ kg m⁻³. Either ρ or χ must be revised for Eq. (2.1) to support $V = 10^9 c$ quantitatively. The postulate $V = 10^9 c$ is therefore used here as an independent assumption.
2. **Casimir constant.** $R_s + R = 7.7665 \times 10^{25} + 0.0039228 \times 10^{25} = 7.7704 \times 10^{25}$ m, not 7.8057×10^{25} m as in the source. Consequently $c_2 = -4.5373 \times 10^{31}$ m² s⁻², $\alpha_0 = -6.1522 \times 10^{45}$ m² and $\alpha_2 = -5.0484 \times 10^{-4}$. The corrections are 0.46 %.
3. **Superluminal factor.** The source quotes "$\beta = i \cdot 2.93 \times 10^{36}$". For a real $V = 10^9 c$, however, $\beta = 10^9$ is real and it is γ that becomes imaginary, Eq. (1.2). The origin of the value 2.93×10^{36} is not given in the source.
4. **Regularity.** The choice $C_2 = 0$ of Part III is regular at infinity but not at $r = 0$. The strictly regular spectrum is given in §III.5.
5. **Casimir pressure.** Eq. (V.2) is a scaling estimate. A regularized mode sum (e.g. by ζ-function) is needed to obtain a finite numerical pressure.

8. Conclusions

The Synchronous Field Theory treats gamma radiation as a longitudinal compression wave of an almost incompressible quantum space, propagating at $V = 10^9 c$. In the generalized Lorentz–Galilei transformation this regime corresponds to an imaginary Lorentz factor. Replacing the provisional equation (4.1) with the full spherical gamma equation (4.2) leads to three results:

- an **exact Airy spectrum**, $\omega_n = |a_n|^{3/2} V/(2^{3/2} R)$, with a fundamental period of about 7.5 days (about 2.7 days for the boundary at R);
- an **explicit Casimir solution** on $[2R, 2R_s]$, with $c_2 = -RV^2/(R_s + R) = -4.5373 \times 10^{31}$ m² s⁻²;
- a **longitudinal flux** $\Phi \propto r^{-4}$ and a Casimir pressure enhanced by 10^9 relative to electrodynamics.

The characteristic crossing time of the local universe, a few days, gives the "synchronous" character of the field a quantitative content. The main open issue is the mechanical justification of $V = 10^9 c$ from the compressibility of space (item 1 of Section 7).

References

1. Popescu, V.-M. — *Synchronous Field Theory*, note SynchronFieldTh.docx, 14 Jan. 2026.
2. Popescu, V.-M. — *TauTimp — Gamma-Ray Longitudinal Wave: Physical Foundations and Full Field-Theoretic Derivation*, 2026.
3. Popescu, V.-M. — *PSTc: Primordial time quanta calculus*. https://michaelvio.github.io/PSTc.pdf
4. Popescu, V.-M. — *Gamma-Ray Planck Constant*. https://michaelvio.github.io/PlanGam.pdf
5. Compressibility modulus of space: https://www.degruyter.com/document/doi/10.1051/978-2-7598-2574-5.c034/html
6. Abramowitz, M. & Stegun, I. A. — *Handbook of Mathematical Functions*, §10.4 (Airy functions).
7. Casimir, H. B. G. — *Proc. K. Ned. Akad. Wet.* **51**, 793 (1948).